FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Advanced)-2019

PAPER – 1

TEST DATE: 05-05-2019

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on this section contains* ***TWO (02)*** *paragraphs. Based on each paragraph, there are* ***TWO (02)*** *questions.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Three vectors $\vec{A}$, $\vec{B}$ & $\vec{C}$ are such that

$\left(\vec{A}\times\vec{B}\right)\times\vec{C}-\left(\vec{A}\times\vec{C}\right)\times\vec{B}=\vec{0}$

Then which of the following may be correct.

(A) $\vec{A}\,||\,\vec{B}$

(B) $\vec{B}\,||\,\vec{C}$

(C) $\vec{C}\,||\,\vec{A}$

(D) Either $\vec{A}$ or $\vec{B}$ or $\vec{C}$ is null vector.

2. Two small stones are thrown simultaneously with the same projection speed v_0 from the top of a pole of height h. One of the stone is thrown towards right horizontally and other stone is thrown at an angle α above horizontally towards left. After time t two stones and top point of pole are lying on same line & x component & y component of relative velocity of first stone w.r.t. second stone are u_x & u_y. It is given that $t=\left[\frac{u_y}{u_x}\right]T$; here T is having the dimensions of time

Pick the correct options:

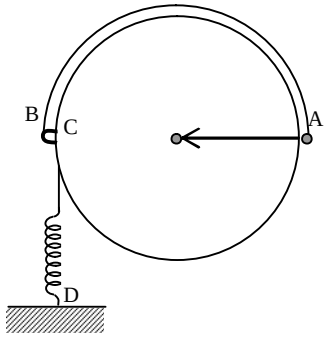
(A) T is independent of α

(B) T is independent of v_0

(C) $\left(\frac{u_y}{u_x}\right)^2\frac{v_0^2}{g}<h<\left(\frac{u_y}{u_x}\right)^2\frac{2v_0^2}{g}$

(D) $\left(\frac{u_y}{u_x}\right)^2\frac{2v_0^2}{g}<h$

3. A cylinder of mass M is placed in such a way that it is free to rotate about its axis. A ball of mass m is attached in one end A of the string and second end of the string is fixed in diametrically opposite end B of the cylinder. A spring of spring constant k is attached at point C (nearest to B) in cylinder whose other end D is fixed on ground. If m < M then pick the correct options. (Initially system is in equilibrium and T is time period)



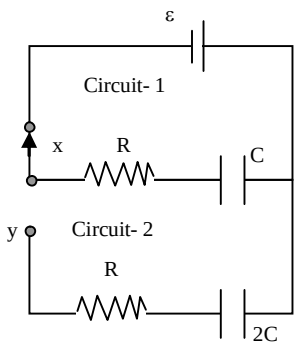
(A) $T<2\pi\sqrt{\frac{3M}{2k}}$

(B) $T>2\pi\sqrt{\frac{3M}{2k}}$

(C) $T>2\pi\sqrt{\frac{3m}{2k}}$

(D) $T<2\pi\sqrt{\frac{3m}{2k}}$

4. A circuit 1 is shown in which a capacitor of capacitance C, a resistor of resistance R and a battery of emf ε are connected together. Switch x can be shifted to position y for changing the connections. After one time constant of circuit 1, key x is shifted to position y at this time, charge on C is Q_1. After one time constant of circuit 2 from this moment charge in capacitor 2C is Q_2. Pick the correct options:



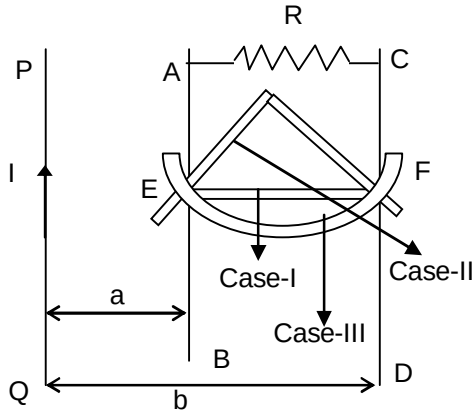
(A) $Q_1 = C\varepsilon\,[1 - e^{-1}]$

(B) $Q_1 = \dfrac{2C\varepsilon[1-e^{-1}]}{3}$

(C) $Q_2 = \dfrac{2}{3}C\varepsilon[1-e^{-1}]$

(D) $Q_2 = \dfrac{2}{3}C\varepsilon[1-e^{-1}]^2$

5. PQ is an infinitely long current carrying conductor AB & CD are smooth conducting rods on which a conductor EF which is a rod in case I an angle in case II& an arc in case III moves with a constant velocity V away from resistance R. Pick the correct options



(A) Maximum force is needed to maintain constant velocity of EF in case III

(B) Force needed to maintain constant velocity of EF is same in all cases.

(C) In case II force needed maintain constant velocity of EF is less than the force needed in case I.

(D) Force needed to maintain constant velocity of EF in case I is given by $F = \dfrac{\mu_0^2 I^2 v}{4R\pi^2}\ln\left(\dfrac{b}{a}\right)$.

6. Pick the correct options:

(A) In Young's double slit experiment, if the width of the source slit is increased, the fringe pattern becomes indistinct.

(B) On refraction energy carried by refracted light is reduced.

(C) Young's double slit experiment can be performed using a source of white light.

(D) In 3D movies 2 projectors are used to project 2 images on screen.

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8

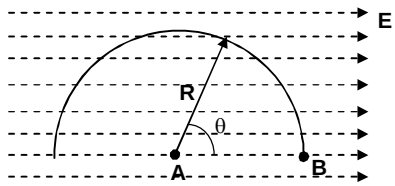
A listener moving with uniform velocity towards a stationary sound source hears a beep of sound of frequency 165 Hz over a distance of 100 m. The actual frequency emitted by the source is 155 Hz and the speed of sound is 330 m/s.

7. Duration of beep emitted by source is
   (A) 4s
   (B) 5s
   (C) 6s
   (D) 7s

8. If listener would move perpendicular to the line joining the listener and stationary source then duration of beep emitted by source would be
   (A) definitely less than time taken in first - case
   (B) definitely more than time taken in first - case
   (C) definitely equal to time taken in first – case exactly
   (D) may be less than or more than time taken in first case.

Paragraph for Question Nos. 9 and 10

A non conducting semi circular ring of radius r having charge distributed over its circumference according to the formula $\lambda = \lambda_0 \sin\theta$ is placed in a uniform electric field E in such a way that plane of the semi circular ring is parallel to electric field E as shown in figure. Consider that polarizability of the semi circular ring is equal to zero.



Answer the following questions.

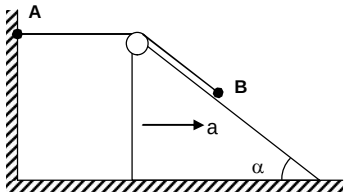
9. Potential at the midpoint of line AB in the absence of electric field E (consider that potential at infinite is zero).
   (A) $\frac{\lambda_0}{\pi\varepsilon_0}$
   (B) $\frac{\lambda_0}{2\pi\varepsilon_0}$
   (C) $\frac{2\lambda_0}{\pi\varepsilon_0}$
   (D) $\frac{\lambda_0}{4\pi\varepsilon_0}$

10. Potential difference between point A and point B
   (A) $\frac{\lambda_0}{2\pi\varepsilon_0} + E.R$
   (B) $\frac{\lambda}{\pi\varepsilon_0} + E.R$
   (C) $\frac{\lambda}{2\pi\varepsilon_0}$
   (D) E.R

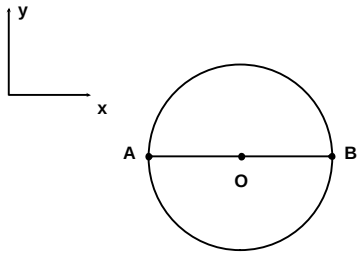
SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A weightless inextensible rope rests on a stationary wedge forming an angle α with the horizontal. One end of the rope is fixed to the wall at point A. A small load is attached to the rope at point B. Wedge start moving to the right with a constant acceleration a and same is the magnitude of acceleration of load. α (in degree) is

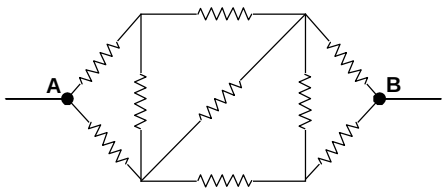


12. A disc of radius 10 cm is placed on a wire parallel to y axis in such a way that it is in equilibrium. AB is the diameter of the disc. Density of the disc linearly increases from A to B starting from zero density at A. Distance of the wire from end A is (in cm)



13. A container of height 96 cm is filled with mercury up to the brim. The container is sealed airtight and a hole of small area n^{th} part of the area of container is punched in its bottom. Time taken upto which mercury come out from the bottom hole is 5 sec. Atmospheric pressure is equal to 76 cm of mercury. n is

14. 9 resistors of resistance 11 Ω are connected in the network shown below. Equivalent resistance between A and B is



15. One train is approaching an observer at rest and another train is receding from him with the same speed 2 m/s. Both trains blow whistles of same frequency 332 Hz. The speed of sound in air is 332 m/s. Beat frequency in Hz as heard by the observer is

16. A biconvex thin lens is prepared from glass of refractive index 3/2. The two bounding surfaces have equal radii of 35 cm each. One of the surfaces is silvered to make it reflecting. An object is placed infront of his lens in such a way that the image coincides with the object. Distance of object from lens is (in cm)

17. The electric potential between a proton and an electron is given by $V = V_0 \ell n\left[\frac{r}{r_0}\right]$, where r_0 is a constant. Assuming Bohr's model to be applicable, the magnetic dipole moment of the electron is proportional to n^x, n is principle quantum number. x is equal to

18. A uniform thin rod of mass 1 kg and length 65 cm is standing on a smooth horizontal table. It slips on the smooth surface and rod starts falling. Velocity of the centre of mass of rod in (m/s) at the instant when it makes an angle 30° with horizontal

Chemistry PART – II

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

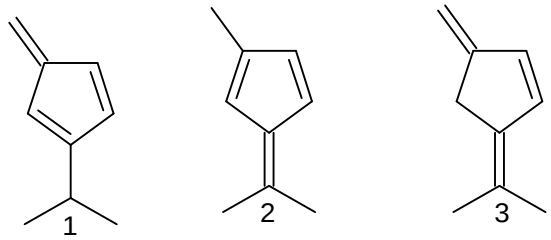
19. (1) $As_2S_3 + 3(NH_4)_2S \longrightarrow 2(NH_4)_3AsS_4$

(2) $Fe(CN)_2 + 4KCN \longrightarrow K_4Fe(CN)_6$

Which of the following option(s) is/are correct?

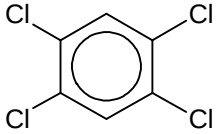
(A) Acids → As_2S_3 KCN
(B) Base → $(NH_4)_2S$, $Fe(CN)_2$
(C) Acids → As_2S_3, $Fe(CN)_2$
(D) Base → $(NH_4)_2S$, KCN

20. Which of the given compound is strongest acid?



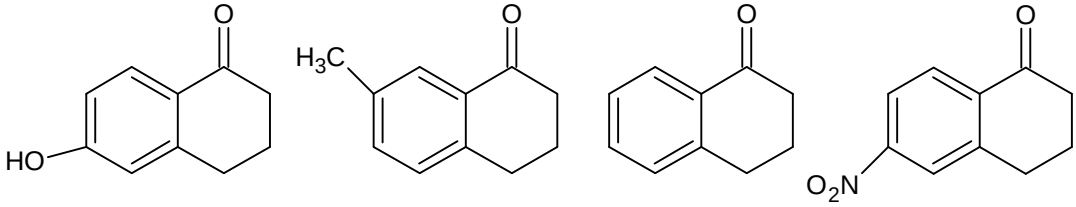
(A) 1
(B) 2
(C) 3
(D) 2 and 3

21. Total number of plane of symmetry present in the given compound is



(A) 3
(B) 4
(C) 5
(D) 6

22. Increasing order of equilibrium constants for the formation of a hydrate:



(i) (ii) (iii) (iv)

(A) iv < iii < ii < i
(B) iv < iii < i < ii
(C) i < ii < iii < iv
(D) ii < iii < i < iv

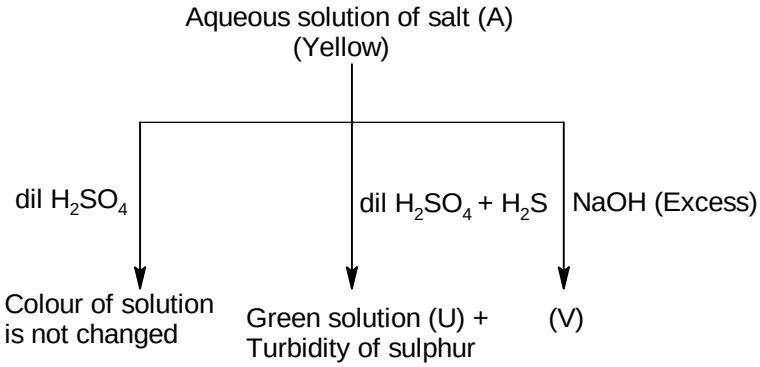
23. Choose the correct options regarding orthoboric acid?
    (A) Orthoboric acid is white crystalline solid with soapy touch.
    (B) Orthoboric acid on strong heating gives B_2O_3.
    (C) It can be prepared by hydrolysis of borax.
    (D) It is monobasic Lewis acid.

24. Select correct statement(s) about the oxoacid $H_4P_2O_6$:
    (A) It is tetrabasic acid
    (B) It has 1 P–O–P linkage
    (C) P atoms show their maximum covalency
    (D) P atoms are in highest oxidation state

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 25 and 26



25. Inorganic salt (A) is
    (A) $CrCl_3$
    (B) K_2CrO_4
    (C) $FeCl_3$
    (D) $AlCl_3$

26. Which observation is correct regarding (V)?
    (A) Brown ppt
    (B) Yellow ppt
    (C) Brown solution
    (D) Yellow solution

Paragraph for Question Nos. 27 and 28

Stereoisomeric complexes have same molecular formula, same constitution they differ only with respect to the spatial orientation of ligand in space around the metal ion.

27. How many geometrical isomers are possible for the complex $[Pt(bn)_2]^{2+}$ [where bn is $H_3C—CH(NH_2)—CH(NH_2)—CH_3$]
    (A) 4
    (B) 5
    (C) 6
    (D) 7

28. How many total stereoisomers are possible for the same complex $[Pt(bn)_2]^{2+}$?
(A) 4
(B) 5
(C) 6
(D) 7

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

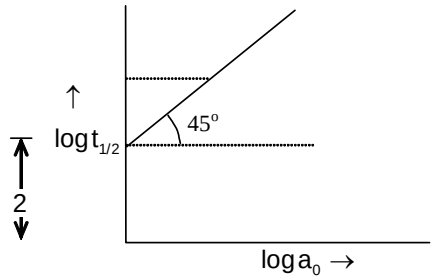
29. A vessel of volume 10 lit. contains 2.8 g of nitrogen gas at a temperature of 1805 K. Find the pressure of the gas (if 50% of its molecules are dissociated into atom at this temperature, [Given: R = 0.082 atm-litre/mol K)

30. 25 ml of a solution of ferric alum $Fe_2(SO_4)_3.(NH_4)_2SO_4.24H_2O$ containing 1.25 gm of the salt was boiled with iron when the reaction $Fe + Fe_2(SO_4)_3 \rightarrow 3FeSO_4$ occurred. The unreacted iron was filtered off and solution treated with 0.107 N $KMnO_4$ in acidic medium. What is volume of $KMnO_4$ solution used?

31. The total number of possible co-ordination isomers of the complex $[Cu^{II}(NH_3)_4]\ [Pt^{II}Cl_4]$ is *x* then value of $\left(\frac{x}{3.2}\right)$ is:

32. The dissociation constant of Acid HA and base BOH at 25°C in aqueous solution is same. The pH of 0.01 M solution of HA is 5. The pH of 0.1 M solution of BOH is

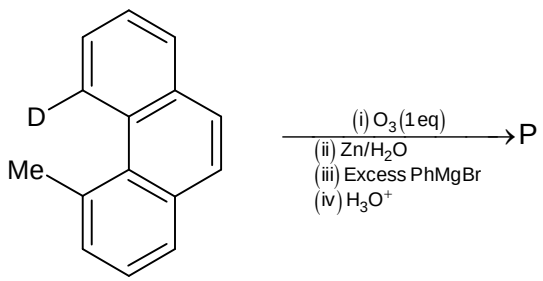
33. In given curve $\left(t_{1/2} \text{ vs } \log a_0\right)$
(where $t_{1/2}$ is half-life and a_0 is initial concentration.)
The value of rate constant is K unit (where unit of K is according as order of reaction). The value of 250 K is:



34. Carbon atoms occupy fcc lattice points as well as alternate tetrahedral voids in diamond. If edge length of the unit cell is 360 pm then the diameter of carbon atom in pm is:

35. A monoatomic gas follow the relation $\frac{P}{V^2} = K$ then molar heat capacity of gas is x, [Assuming there is no vibrational degree of freedom]. What is the value of x in calorie [Given : R = 1.98 cal./mol K]

36. The total number of stereoisomers obtained in the product of the following reaction is *x*. The value of $\frac{x}{5}$ is.



Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $\lim_{x\to 0} \frac{ae^x + b\ln(1+x) + 5}{x^2}$ = L(finite), then
(A) a = 5
(B) b = –5
(C) a + b = 0
(D) b + L – 2a = 10

38. Let $\lim_{n\to\infty} \prod_{k=3}^{n}\left(1-\tan^4\frac{\pi}{2^k}\right) = \frac{\pi^3}{m}$, then 'm' is greater than
(A) 8
(B) 16
(C) 28
(D) 36

39. Let $a_1, a_2, a_3, \ldots, a_n$ are in A.P., and $a_3, a_5, a_8, b_1, b_2, b_3, \ldots b_n$ are in G.P. and $a_9 = 40$, then
(A) $\sum_{i=1}^{9} a_i^2 = 6144$
(B) $\sum_{i=1}^{\infty} \frac{1}{b_i} = \frac{1}{18}$
(C) $\sum_{i=1}^{9} a_i^2 = 6278$
(D) $\sum_{i=1}^{\infty} \frac{1}{b_i} = \frac{1}{28}$

40. Let $A = C_1C_3 + C_2C_4 + C_3C_5 + C_4C_6 + \ldots + C_{m-2}C_m$, (where $C_r = {}^mC_r$), then
(A) $A \geq {}^{2m}C_{m-2}$
(B) $A < {}^{2m}C_{m-2}$
(C) $A > \sum_{r=0}^{m} C_r^2$
(D) $A < \sum_{r=0}^{m} C_r^2$

41. Let number of values of 'r' for which ${}^{15}C_r$ is an odd integer, be 'k', then k is greater than
(A) 7
(B) 11
(C) 9
(D) 5

42. Let $A(z_1)$, $B(z_2)$, $C(z_3)$ lies on a circle $|z| = r$ and forms an acute angled triangle, and $\omega = \frac{z + z_1}{z_2 - z_3}$, where P(z) is the point where altitude from A to BC meets the circumcircle, then
(A) ω is purely real
(B) ω is purely imaginary
(C) $\text{Arg}(\omega) = 0$
(D) $\text{Arg}(\omega) = \pi$

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

Let $(x^3 + x^5)(2x^6 + 3x^4 - 1)^{20} = a_0 + a_1x + a_2x^2 + ... + a_{125}x^{125}$

43. The $a_0 - \frac{a_2}{3} + \frac{a_4}{5} - \frac{a_6}{7} ..... + \frac{a_{124}}{125}$ is equal to
(A) 2^{41}
(B) $\frac{2^{40}}{63}$
(C) $\frac{-1}{256}$
(D) none of these

44. Which of the following is incorrect?
(A) $a_0 = a_2 = a_4 ..... = a_{124}$
(B) $a_{125} = 2^{20}$
(C) $a_0 + a_1 + a_2 = 0$
(D) $a_1 + a_2 + a_3 = 0$

Paragraph for Question Nos. 45 and 46

Let $I(f(x)) = \int_0^{2\pi} \frac{2^{f(x)}}{2^{f(x)} + 2^{-f(x)}} dx$, where f(x) is real valued function

45. The value of $I\left(\frac{1}{\ln 2}\right)$ is equal to
(A) $\frac{2\pi 2^{2e}}{2^{2e} + 1}$
(B) $\frac{\pi \cdot 2^{2e}}{2^{2e} + 1}$
(C) $\frac{2\pi e^2}{e^2 + 1}$
(D) $\frac{\pi e^2}{e^2 + 1}$

46. The value of I(sin x) is equal to

(A) $\frac{\pi}{2}$

(B) π

(C) 2π

(D) 4π

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. In an ellipse $\frac{x^2}{10}+\frac{y^2}{6}=1$. If focal chord PSP′ and QSQ′ are at right angles to each other, then $\frac{1-e^2}{(SP)(SP')}+\frac{1-e^2}{(SQ)(SQ')}=\frac{N}{15}$, then value of [N] (where [.] denotes the greatest integer function)

48. If $\frac{\left(1^4+\frac{1}{4}\right)\left(3^4+\frac{1}{4}\right).....\left((2n-1)^4+\frac{1}{4}\right)}{\left(2^4+\frac{1}{4}\right)\left(4^4+\frac{1}{4}\right).....\left((2n)^4+\frac{1}{4}\right)}=\frac{1}{k_1n^2+k_2n+k_3}$, then $k_1 - k_2 + k_3$ equal to

49. Let (x, y) be a pair of real numbers satisfying $56x+33y=\frac{-y}{x^2+y^2}$ and $33x-56y=\frac{x}{x^2+y^2}$. If $|x|+|y|=\frac{p}{q}$ (where p and q are relatively prime), then (6p – q) is

50. The equation $x^4 + ax^3 + bx^2 + ax + 1 = 0$ has atleast one real root and if the minimum value of $E(a, b) = a^2 + b^2$ can be expressed as $\frac{p}{q}$ (p and q are relatively prime), then $(p^2 + q^2)$ is

51. A bag contains (2n + 1) coins. It is known that 'n' of these coins have tail on both sides, whereas the remaining (n + 1) coins are fair. A coin is randomly drawn from bag and tossed. If the probability of that the toss results in a tail is $\frac{31}{42}$, then value of n is

52. Let $\Delta=\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}$, where D_1, D_2, D_3 are cofactors of c_1, c_2 and c_3 respectively such that $D_1^2+D_2^2+D_3^2=16$ and $c_1^2+c_2^2+c_3^2=4$, then maximum value of Δ is

53. Let the value of $\int_1^{\sqrt{2}}(x^4-2x^2+1)^{\frac{1}{3}}dx+\int_{-1}^{0}\sqrt{(x+1)^{\frac{3}{2}}+1}dx$ be I, then [I] is equal to (where [.] denotes the greatest integer function)

54. Let $\int\frac{2e^{5x}+e^{4x}-4e^{3x}+4e^{2x}+2e^x}{(e^{2x}+4)(e^{2x}-1)^2}dx=\tan^{-1}\left(\frac{e^x}{a}\right)-\frac{1}{b(e^{2x}-1)}+c$ (where a, b, c are constants), then the value of a + b is equal to